23 Testing Settlement–Mobility Relationships with “From Everywhere To Everywhere” Least-Cost Paths and Monte Carlo Simulations: a case study from Hellenistic Armenia

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Understanding how ancient settlements related to communication axes is essential to interpreting access to resources, networks of exchange and strategies of territorial control. Since limited evidence for archaeologically attested ancient routes often makes these relationships difficult to assess, GIS-based modelling and spatial statistics offer a means of exploring how landscapes channel movement along topographically favourable corridors and how settlement patterns relate to them. To move beyond “eyeballing” distribution maps, this paper combines Least-Cost Paths “From Everywhere To Everywhere” (LCP FETE) with Monte Carlo spatial randomisation in a reproducible FLOSS workflow implemented using QGIS, GRASS and R. Unlike site-to-site least-cost analyses, FETE models concentrations of potential movement without using known settlements as origins or destinations, thereby reducing circularity between the evidence to be explained and the explanatory model. Through a Monte Carlo procedure, the proximity of archaeological sites to high-density modelled corridors is then compared with that of simulated site distributions under an environmentally constrained null model. Tests are stratified by chronology and settlement type, while the effects of modelling choices and computational constraints are critically considered. The approach is illustrated through its application to the Hellenistic-period settlement pattern of the southern basin of Lake Sevan,in Armenia. This resource-rich highland landscape, historically crossed by important trade and caravan routes, offers a rural perspective complementary to research centred on Armenia’s royal capitals and monumental remains, while also providing a suitable setting for testing the proposed methodology. Drawing on legacy data from the Armenian–Italian YSU–CNR mission, the workflow examines how fortified and open settlements relate to modelled movement patterns, also contributing to a broader understanding of rural territorial organisation during the formation and consolidation of the first independent Armenian kingdom